



Acute coronary syndrome prediction in emergency care: A machine learning approach

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ABSTRACT

Background and Objective: Clinical concern for acute coronary syndrome (ACS) is one of emergency medicine's most common patient encounters. This study aims to develop an ensemble learning-driven framework as a diagnostic support tool to prevent misdiagnosis.

Methods: We obtained extensive clinical electronic health data on patient encounters with clinical concerns for ACS from a large urban emergency department (ED) between January 2017 and August 2020. We applied an analytical framework equipped with many well-developed algorithms to improve the data quality by addressing missing values, dimensionality reduction, and data imbalance. We trained ensemble learning algorithms to classify patients with ACS or non-ACS etiologies of their symptoms. We used performance evaluation metrics such as accuracy, sensitivity, precision, F1-score, and the area under the receiver operating characteristic (AUROC) to measure the model's performance.

Results: The analysis included 31,228 patients, of whom 563 (1.8%) had ACS and 30,665 (98.2%) had alternative diagnoses. Eleven features, including systolic blood pressure, brain natriuretic peptide, chronic heart disease, coronary artery disease, creatinine, glucose, heart attack, heart rate, nephrotic syndrome, red cell distribution width, and troponin level, are reported as significantly contributing risk factors. The proposed framework successfully classifies these cohorts with sensitivity and AUROC as high as 86.3% and 93.3%. Our proposed model's accuracy, precision, specificity, Matthew's correlation coefficient, and F1-score were 85.7%, 86.3%, 93%, 80%, and 86.3%, respectively.

Conclusion: Our proposed framework can identify early patients with ACS through further refinement and validation.

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1. Introduction

Acute coronary syndrome (ACS) is comprised of a spectrum of diseases categorized by three distinct processes, each with its own diagnostic and therapeutic challenges: [1] (i) *ST-segment elevation myocardial infarction (STEMI)*, in which myocardial ischemia results from total coronary artery occlusion, (ii) *Non-ST-segment elevation myocardial infarction (NSTEMI)*, in which myocardial ischemia results from partial occlusion, and (iii) *Unstable angina (UA)*, in which there is dynamic plaque occlusion of the coronary artery placing patients at high-risk for ischemia.

Acute coronary syndrome is a leading cause of morbidity and mortality in developed and developing countries [2]. Clinically, patients with ACS complain of chest pain, occasional radiation to the arm, jaw, or shoulder, lightheadedness, shortness of breath, nausea, vomiting, or dyspnea [3]. While these symptoms are common in emergency department (ED) patients, the diagnosis of ACS remains challenging [4]. The need to accurately rule out ACS in patients leads to substantial over-testing that is costly and time-consuming. [5]

In the past, overestimating the potential for these conditions has been the standard clinical practice in the emergency department, as increased false-negative results can lead to life-threatening situations. However, unnecessary tests such as electrocardiography (ECG), cardiac enzyme tests, and long-term hospitalization of patients with chest pain increase the burden of illness [5]. The clinical diagnosis in the ED of ACS relies on measuring

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cardiac markers, electrocardiogram abnormalities, and patient history [6]. Precise identification of patients with coronary artery occlusion or near-occlusion who will benefit from early revascularization can reduce mortality, and accurate identification of those with noncardiac causes of their symptoms can substantially reduce costs and resource utilization. Several risk scores have been developed for assessing mortality risk in patients with ACS, including the Global Registry of Acute Coronary Events (GRACE) risk score [7], Acute Coronary Treatment and Intervention Outcomes Network (ACTION) risk score [8], Thrombolysis in Myocardial Infarction (TIMI) risk score, Canadian ACS risk score [9], and Emergency Department Assessment of Chest Pain Score (EDACS) [10]. While existing risk scoring systems have improved the identification of ED patients at high-risk for ACS from noncardiac etiologies, they sacrifice specificity for high sensitivity.

In their study, Huang, Ying [11], stated that In addition to traditional predictors, self-reported physical activity, average daily heart rate, fluctuations in arousal blood pressure, and percentage time of diastolic hypertension contributed significantly to the risk classification of cardiovascular disease. In a study by [12], they proposed an ensemble that outperforms individual and ensemble classifiers in all three datasets containing different feature spaces. In other words, its generalization ability is excellent. Also, the costs associated with missing patients (false negatives) are much higher than those associated with mislabeling (false positives) in healthy patients. Given the various misclassification costs, the proposed method will be more realistic. They also combined the accuracy and misclassification cost as metrics to allow an ensemble classifier to improve accuracy while reducing misclassification cost. According to [13], their study proposed a method that provides a prediction system that detects and critically analyzes vital risk factors in high-dimensional health data to accurately predict heart disease before a heart attack or stroke occurs. The proposed ensemble deep learning model effectively processes two different data sources to improve the performance of heart disease diagnosis. This framework identifies the best features and calculates their particular importance within the dataset to improve the accuracy of the predictions.

Mienye, Sun [14] conducted a comparative study using the Cleveland and Framingham datasets and conducted it with some recent scientific papers and other well-known machine learning algorithms such as k-nearest neighbors (KNN), logistic regression (LR), linear discriminant analysis (LDA), support vector machines (SVM), Classification and Regression Tree (CART), Gradient Boosting and Random Forest. The experimental results on the dataset achieved classification accuracies of 93% and 91%, respectively, which outperformed other machine learning algorithms.

Unbalanced datasets present significant difficulties in terms of supervised classification. Noncardiac etiologies for chest pain are far more common than ACS, thus leading to an unbalanced dataset for analysis. Consequently, learning from an unbalanced dataset is difficult in various practical machine learning (ML) applications [15]. In addition, class imbalance requires adopting imbalanced analytical approaches such as sampling, machine learning, and ensemble learning [16]. Moreover, although extensive classification studies using unbalanced datasets have been conducted by the ML and data mining community [17], little or no multi-class ML classifiers have been developed for heart disease applications.

Preprocessing techniques are classified into three groups: (i) *Undersampling techniques* produce a subset of the initial dataset by eliminating certain instances of the majority group. (ii) *Oversampling techniques*: these produce a superset of the initial dataset by either duplicating the instances of the minority class or creating new ones using the original minority class instances. (iii) *Hybrid methods (Resampling techniques)* combine the undersampling and the oversampling techniques. They are also sensitive to over-

fitting issues; they can dispose of minority class instances by applying oversampling methods to address overfitting problems [18]. Data modeling techniques to account for data imbalance problems significantly impact the accuracy of results [19]. The previously conducted studies showed that sampling techniques and ML algorithms best handle data imbalance streams [20].

A key focus of this study is modeling highly unbalanced tasks through improved artificial learning techniques. We use a multi-step analysis strategy for modeling imbalanced classification problems in stratifying patients with clinical concerns for ACS. This strategy includes an initial step model of balanced datasets and a subsequent step that applies and compares ML classifiers and their evaluation performance metrics. This study analyzed the multi-class classification problem using ensemble learning techniques. The fundamental rationale for choosing this approach is that they are both simple to decipher and effective in solving class-imbalance issues [21]. The reality of ACS is that with STEMI, it typically has a straightforward EKG-based diagnosis and does not require further decision support, unlike its other counterparts (i.e., NSTEMI and UA), which involves decision support. As a result, our primary objective was to develop an ensemble, ML multi-class framework to classify suspected ACS patients into those with ACS (NSTEMI and UA) or non-ACS etiologies.

2. Methods

The primary purpose of this research is to propose a framework to classify and distinguish NSTEMI and UA from non-ACS etiologies in the ED. The proposed framework is built on five phases. The first phase, data preprocessing, involves data cleaning, imputation, transformation, and standardization. The next phase, feature selection, reduces the dimension of the dataset by identifying the most suitable features. This leads to the development of a more reliable model, minimizing the risk of overfitting, enhancing accuracy, and reducing the time to train. The third phase handles the data imbalance by utilizing an integration of cost-sensitive classification and resampling techniques. Next, an 80:20 percent train test split (i.e., 80% training vs. 20%testing) is performed to evaluate the performance of the trained ensemble learning models (i.e., Adaptive Boost (AdaBoost), Gradient Boosting, and Extreme Gradient Boosting (XGBoost)) based on several evaluation metrics such as the AUROC, sensitivity, precision, and F1-score in a robust manner. The ground truth labels (i.e., NSTEMI, UA, and non-ACS etiologies) were established through a panel of adjudicators which is a mix of cardiologists and emergency physicians.

2.1. Data description

During the study period, we used a private dataset comprising 362,138 patients who visited a large urban ED in the US. The data records include demographic information, laboratory values, comorbidities, vital signs, and primary diagnoses coded using the International Classification of Diseases Clinical Modification, 10th Revision (ICD-10-CM). The exclusion criteria for this study consist of STEMI, transfer from another hospital, and primary residence outside the state of Michigan. Fig. 2 displays the exclusion criteria of this cohort.

Of the initial 362,138 patients that visited the ED, 330,612 patients are excluded due to unrelated complaints, and 31,526 patients with ACS indications are assessed. 308 STEMI patients are further excluded from our study due to our focus on NSTEMI, UA, and non-ACS etiologies. Conclusively, a total of 31,228 patients with clinical concerns for ACS are selected for this study, of which patients with NSTEMI, UA, and non-ACS etiologies are 431 (1.4%), 132 (0.4%), and 30,665 (98.2%), respectively (Fig. 2).

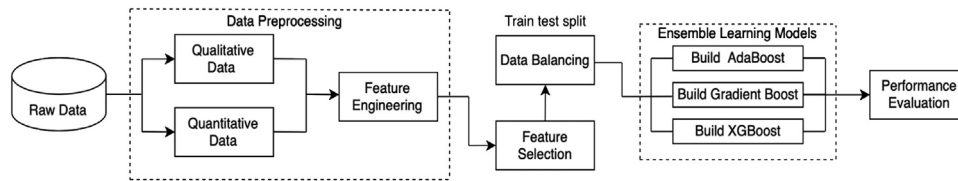


Fig. 1. Architecture of proposed methodology.

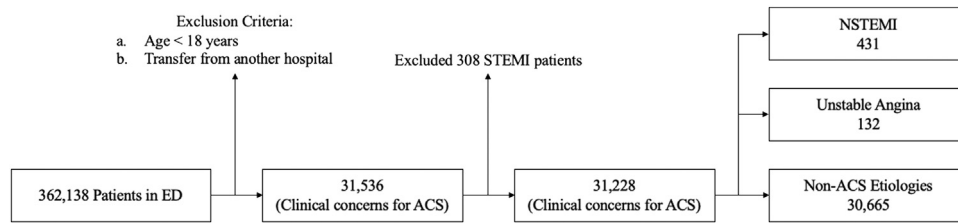


Fig. 2. Selection process for the study cohort.

We understand that the prevalence of ACS accounts for only about 1.8% of our data. The values are low because, from the data provided, we are only considering the lower-risk patients without an initial instance of high troponin levels. These patients do not immediately rule out NSTEMI or UA by their initial troponin values. So, therefore, by definition, we are dealing with a lower acuity group of patients.

2.2. Data preprocessing

We identified a broad list of potential clinical features associated with ACS through a review of the medical literature. From these lists, we selected 58 variables routinely recorded in the electronic health record, including demographic and clinical factors, to include in the analysis.

2.2.1. Feature engineering

This process involves utilizing domain knowledge to extract features from a record set/dataset. The following stages are performed during our analysis to achieve the intended objective: (i) *One-hot encoding*: replacement of encoded categorical variables with newly generated binary variables to facilitate better classification, as displayed in Fig. 1. (ii) *Removing redundant/duplicate data*: noise, missing values, and discrepancies capable of adversely impacting the dataset's quality and altering the prediction model are present in our raw dataset. Therefore, duplicate data rows and features having more than 70% missing values are removed. (iii) *Data imputation*: through Multiple Imputations by Chained Equations (MICE) [22].

2.3. Feature selection

Feature Selection is the process of streamlining features to only those contributing the most to the output [23]. BorutaShap is a wrapper feature selection technique that consolidates the Boruta [24] feature selection algorithm with the SHAP (SHapely Additive exPlanations) values [25]. BorutaShap is applied when identifying the variables with the most contribution to enhancing classification performance [20].

2.4. Data balancing (Resampling techniques & cost-sensitive classification)

A sequential integration of resampling techniques (i.e., the combination of synthetic minority oversampling technique (SMOTE) and Tomek links (i.e., SMOTETomek)) and cost-sensitive classification is applied to address the data imbalance. Tomek Links is a

technique for identifying pairs of nearest neighbors in a dataset with different classes. In implementing the resampling techniques, the idea is to balance the dataset with SMOTE and remove Tomek links from all classes. This implies that we apply Tomek links to the oversampled training dataset as a data cleaning approach. Subsequently, we have incorporated the cost-sensitive classification, a multi-class classification problem, defined as an associated vector of costs for labeling each observation under each label. The aim is to build a classifier that predicts the class with the minimum expected cost. *Cost-sensitive classification* incorporates different costs of prediction errors into the algorithm when training an ML model [26]. A cost matrix for each class is defined using the costsensitive package available in Python to achieve this. It assigns a different cost for different types of classification errors and features of each observation. At the same time, resampling techniques balance the data by removing the majority of class instances that form Tomek links and eliminating instances created by SMOTE [27].

2.5. Ensemble learning models

Ensemble learning methods are developed by aggregating the classification results of multiple learning algorithms instead of relying upon a single model [28,29]. These models promise better performance than individually trained ensemble models and provide a robust classification due to reducing the variance of the classification error [59]. This study implements three boosting ensemble models with the proposed framework. a) AdaBoost (i.e., Adaptive Boosting) tweaks subsequent weak learners to build a robust classifier. Although individual learners can be weak, the final model can be proven to converge into a strong learner [30]. b) Gradient Boosting is an ML technique for regression, classification, and other tasks, which yields a prediction model in the form of an ensemble of weak prediction models, usually decision trees [31,32]. Once a decision tree is a weak learner, the subsequent algorithm is called Gradient boosted trees, outperforming random forest. c) XG-Boost, also known as Extreme Gradient Boosting, is a widely used ML algorithm for classification and regression problems. It implements Gradient boosted decision trees designed for speed and performance [33].

2.6. Model evaluation

The confusion matrix is used to summarize the model's performance. The performance metrics used in this analysis are as follows:

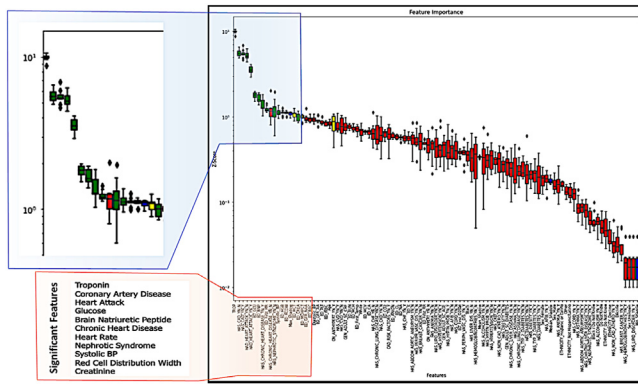


Fig. 3. Selection of potential variables by BorutaShap.

- **True Positive (TP) and True Negative (TN):** A true positive occurs where the model effectively predicts the positive class. Likewise, a true negative occurs where the model accurately predicts the negative class.
- **False Positive (FP) and False Negative (FN):** A false positive occurs where the model ineffectively predicts the positive class. Likewise, a false negative occurs where the model inaccurately predicts the negative class.
- **Accuracy:** The accuracy of a model is characterized as the total number of patients with heart disease divided by the overall number of patients.
- **Precision:** Precision is used to decide how many have heart disease out of the number of patients predicted to have heart disease.
- **Sensitivity:** Sensitivity is used to decide how many heart disease patients got a positive test result out of the number of heart disease patients.
- **Specificity:** Specificity is used to decide, out of the number of patients who do not have heart disease, how many got a negative test result.
- **Matthew's Correlation Coefficient (MCC):** It signifies the prediction ability with values between $[-1, +1]$. If the estimation of the MCC classifier is $+1$, this implies the classifier predictions are ideal. -1 demonstrates that the classifiers produced wrong predictions. It is characterized as

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

3. Results

Table 1 shows the baseline characteristics of our dataset (i.e., patients with NSTEMI, UA, or non-ACS etiologies), where all measurable tests have a significance level of 5%. The age 64.36 ± 2.03 represents a mean of 64.36 and a standard deviation of 2.03. The dominance of atrial fibrillation (37% vs 7.6% vs 7.4%), congestive heart failure (31% vs 28% vs 19%), chronic heart failure (37% vs 34% vs 18%), chronic kidney disease (30% vs 22% vs 25%), cardiovascular disease (58% vs 56% vs 33%), heart attack (35% vs 13% vs 4.7%), and obesity (17% vs 14% vs 15%) are observed more dominantly in NSTEMI patients is represented in Table 1.

Fig. 3 shows the contributions of each feature towards prediction through BorutaShap, where the green, yellow, and red bars represent accepted, tentative, and rejected features, respectively. A total of 11 features, including systolic blood pressure, brain natriuretic peptide (BNP), chronic heart failure (CHF), coronary artery disease (CAD), creatinine, glucose, history of myocardial infarction (i.e., heart attack), heart rate, nephrotic syndrome, red cell distribu-

tion width (RCDW), and troponin level, are reported as significantly contributing risk factors.

For each identified feature, a set of ranges is defined based on the third quartile (i.e., the 75th percentile) and the maximum. Table 2 shows the selected features and their range of values.

Fig. 4 shows the data imbalance before and after implementing the resampling technique. Before the data imbalance, NSTEMI, UA, and non-ACS etiologies are 345, 97, and 24,540 for the training set. After data balancing, NSTEMI, UA, and non-ACS etiologies' training set becomes 24,540, 24,540, and 24,528, respectively.

Fig. 4 shows that the dataset became balanced after implementing the resampling technique on the training dataset (SMOTE-Tomek). The only downside with this is that, with the application of SMOTE, synthetic samples are generated for the minority classes, which are not as concrete as the majority class, leading to low precision, sensitivity, and F1-score values. To further address this issue, we implement the cost-sensitive classification approach. Hence, the *costsensitive* package in Python, as discussed in Section 2, helps determine the optimal cost matrix required to build a classifier that predicts the class with the minimum expected cost. Table 3 shows the total misclassification costs and associated accuracies based on the boosting ensemble models.

From Table 3, we can conclude that the higher the misclassification cost, the lower the accuracy. We can justify this statement because, in cost-sensitive learning, each instance is assigned a misclassification cost instead of correctly or incorrectly classified. Thus, the problem is to minimize the total misclassification cost instead of optimizing the accuracy.

The proposed framework uses SHAP values to extract feature importance. When a classifier makes a specific prediction based on a set of features, it calculates the performance change with and without the presence of each feature. Those features leading to a significant performance reduction with their absence will be assigned a higher contribution score. Fig. 5 shows the SHAP values attributed to the boosting ensemble learning used. Each point represents a sample in training set along with their SHAP scores. The variables are ranked in descending order, i.e., the troponin level has the highest feature importance, and a history of chronic heart disease has the lowest importance. The horizontal location depicts whether the effect of that value is associated with either a high or low predictive value. For instance, for the XGBoost model, a high "troponin" level has a high and positive impact on the type of primary diagnosis for the patients. The "high" stems from the red color, and the "positive" impact is depicted from the horizontal axis. Similarly, we can conclude that "heart rate" negatively correlates with the target variable.

From our proposed approach, we can see that the Gradient Boosting model outperformed the other ensemble models. The AU-ROC curve for the three boosting ensemble models is illustrated in Fig. 6.

4. Discussion

To study the significance of different steps in our proposed algorithm, we trained predictive models on raw data, transformed data via cost-sensitive classification and feature selection, and transformed data via cost-sensitive classification and feature selection and resampling, as displayed in Table 4. Table 4 shows the performance comparison of the models used at different levels of data processing. Although all models display a relatively high average accuracy score of 97.7% on raw data, they lack precision and sensitivity, leading to a very low F1-score due to low precision and sensitivity. This is caused by the data imbalance where the models learn classification of every data point as non-ACS, providing an average accuracy of 97.7% as the dataset is heavily biased toward the non-ACS patients given that 30,655 out of 31,228 patients (i.e.,

Table 1
Baseline characteristics of selected features.

Variables	NSTEMI(N = 431)	UA(N = 132)	Non-ACS(N = 30,665)	p-value
Age	64.36± 2.03	60.05± 0.62	59.11± 0.01	<0.001
Sex				
Female (%)	48	42	53	N/A
Male (%)	52	58	47	
Ethnicity				
Decline (%)	2	–	2	N/A
Do not know (%)	1	–	1	
Hispanic/Latino (%)	3	4	5	
Not Hispanic/Latino (%)	94	96	92	
On treatment for hypertension (%)	65	73	51	
Abdominal aortic aneurysm (%)	1	2	1	
Atrial fibrillation (%)	37	8	7	
Anemia (%)	20	18	21	
Asthma (%)	5	8	10	
Breast cancer (%)	1	–	1	
Coronary artery disease (%)	54	61	13	
Congestive heart failure (%)	31	28	19	
Chronic heart failure (%)	37	34	18	
Chronic liver disease (%)	5	8	7	
Chronic lung disease (%)	20	24	22	
Chronic kidney disease (%)	30	22	26	
Chronic obstructive pulmonary disease (%)	15	19	15	
Cardiovascular disease (%)	58	57	33	
Diabetes (%)	30	38	25	
Type-2 diabetes (%)	32	40	26	
Hypertension (%)	60	61	48	
History of hemoglobinopathies (%)	–	–	–	
Lipid disorder (%)	–	–	–	
Nephrotic syndrome (%)	1	–	–	
Noncardiac atherosclerosis (%)	–	–	–	
Peripheral vascular disease (%)	11	11	5	
Heart attack (%)	35	13	5	
Liver disease (%)	69	78	72	
CAD risk factors (%)	68	76	58	
Obesity (%)	17	14	15	
Systolic blood pressure	147± 1.85	140± 6.03	138± 0.03	<0.001
Diastolic blood pressure	85± 0.83	79± 2.71	81± 0.01	<0.001
Heart rate	87± 1.01	77± 3.29	88± 0.01	<0.001
Temperature	98± 0.01	98± 0.03	98± 0.00	0.847
Respiratory rate	20± 0.07	19± 0.22	20± 0.00	0.021
Oxygen saturation	97.03± 0.05	97.93± 0.15	97.22± 0.00	0.121
Troponin	2.39± 0.01	0.06± 0.02	0.10± 0.00	0.000
Brain natriuretic peptide	610.43± 1873.44	160.71± 6238.75	397.61± 35.87	<0.001
Red blood cell	4.38± 0.00	4.44± 0.00	4.29± 0.00	0.005
White blood cell	8.96± 0.10	7.25± 0.30	8.40± 0.00	0.019
Platelet count	234.27± 61.92	224.74± 19.28	232.03± 0.27	0.246
Hematocrit	38.48± 0.10	38.53± 0.31	37.46± 0.00	<0.001
Hemoglobin	12.36± 0.01	12.78± 0.04	12.31± 0.00	<0.001
Mean corpuscular hemoglobin	29.00± 0.0	28.90± 0.07	28.85± 0.00	0.615
Mean corpuscular hemoglobin concentration	32.84± 0.00	33.10± 0.01	32.81± 0.00	0.015
Mean corpuscular volume	88.21± 0.14	87.25± 0.46	87.80± 0.00	0.393
Red cell distribution width	15.38± 0.02	15.04± 0.05	15.56± 0.00	0.022
Anion	9.11± 0.11	10.47± 0.04	9.73± 0.00	<0.001
Blood urea nitrogen	9.73± 0.00	23.69± 0.00	21.62± 0.01	0.022
Calcium	9.30± 0.00	9.31± 0.00	9.27± 0.00	0.418
Glucose	137.50± 77.98	164.92± 24.16	134.73± 0.34	<0.001
Chlorine	103.11± 0.06	104.21± 0.06	102.87± 0.00	0.009
Carbon dioxide	23.52± 0.04	24.08± 0.13	24.35± 0.00	<0.001
Creatinine	1.82± 0.01	1.36± 0.03	1.60± 0.00	0.039
Potassium	4.22± 0.00	4.08± 0.01	4.14± 0.00	0.104
Sodium	137.09± 0.04	137.39± 0.13	139.95± 0.00	0.374
Glomerular filtration rate	69.72± 2.82	81.16± 9.24	78.49± 0.04	<0.001
Glomerular filtration rate Non-African American	60.37± 2.4	70.48± 7.69	69.03± 0.03	<0.001

98.2%) are suffering from non-ACS. This leads to significant false positive and false negative classifications and results in low precision (i.e., 47.7%), sensitivity (i.e., 43.0%), F1-score (i.e., 43.3%), and MCC (i.e., 13.7%). Although we had good specificity (i.e., 74.0%). where: CSC – Cost-sensitive classification, FS – Feature selection, and RS – Resampling

As shown in Table 4, feature selection and cost-sensitive classification combined led to a minor increase in the average accuracy,

precision, MCC, and F1-scores by 0.3%, 0.3%, 1.0%, and 0.4%, respectively. It only reduced the sensitivity and specificity scores by 1.3% and 0.7%, respectively. Finally, our proposed model (i.e., the combination of feature selection, cost-sensitive classification, and resampling) does reduce the average accuracy score by 12.2%. Still, it also improves the average precision, sensitivity, specificity, MCC, and F1-scores by 38.6%, 43.3%, 19%, 66.3%, and 43%, respectively. Also, the MCC scores for the raw data and transformed data via

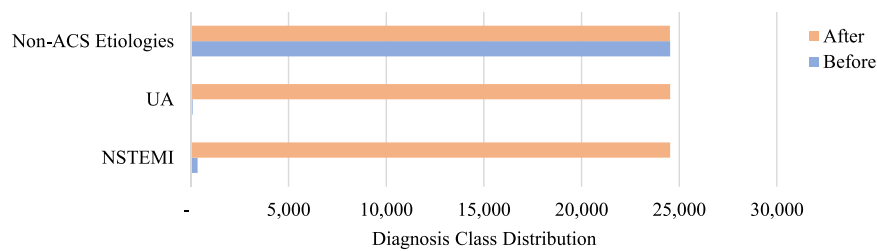


Fig. 4. Data distribution of the training set before and after resampling.

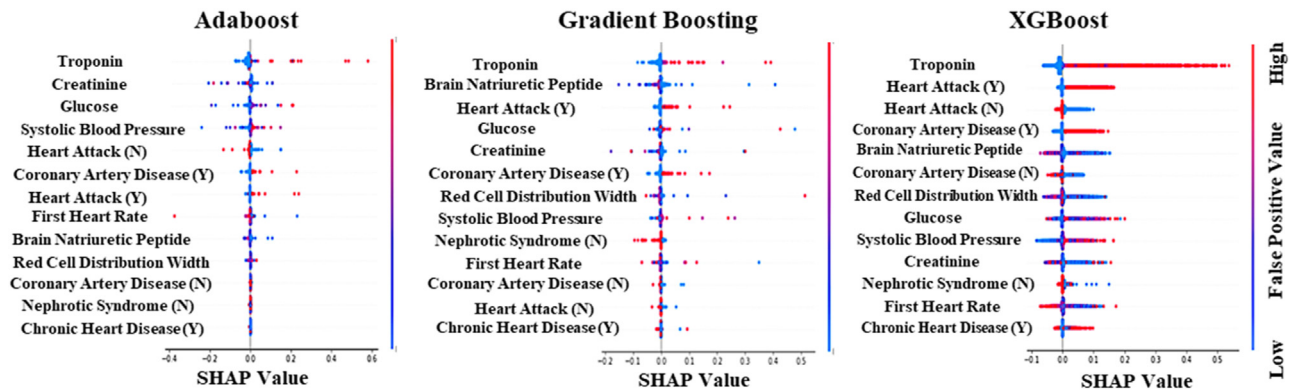


Fig. 5. SHAP values for Adaboost, Gradient Boosting, and XGBoost, respectively.

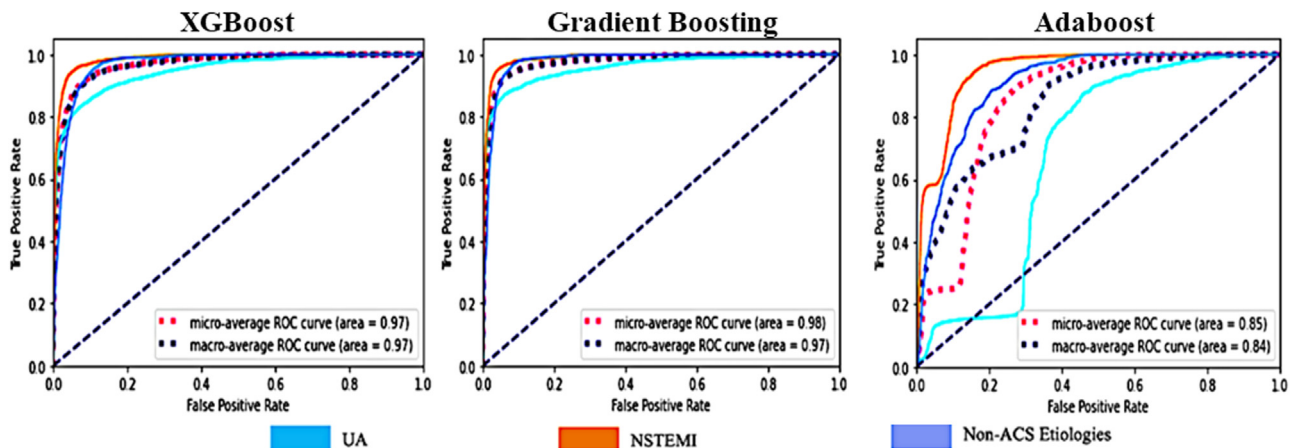


Fig. 6. ROC for XGBoost, Gradient Boosting, and AdaBoost, respectively.

Table 2

Range of continuous variables independently associated with NSTEMI.

Features	Range of Values	Unit
Brain Natriuretic Peptide	200 – 555	pg/mL
Creatinine	1.3 – 1.8	mg/dL
Glucose	160 – 225	mg/dL
Heart Rate	95 – 137	beats per minute
Red Cell Distribution Width	15.6 – 18.6	%
Systolic Blood Pressure	130 – 210	mmHg
Troponin	0.4 – 1.00	ng/mL

Table 3

Misclassification cost and accuracy for the chosen cost matrix.

Boosting Models	Total Misclassification Cost	Accuracy (%)
AdaBoost (ADB)	7683.60	79
Gradient Boosting (GRB)	3521.35	90
XGBoost (XGB)	4586.46	88

the combination of FS + CSC based on the XGBoost model have no record due to its negative value.

Since we are initially dealing with a framework of imbalanced datasets, accuracy automatically becomes an improper measure since it cannot distinguish between the numbers of correctly classified examples of different classes. Thus, measures like F1-score become more favorable in this case, as the False Negatives and

False Positives are crucial. Also, given that we applied an oversampling technique to our data, it puts more weight on the smaller classes, making the model biased. The model will now predict the smaller classes with higher accuracy, but the overall accuracy will decrease.

4.1. Main findings

Our current study findings show that the Gradient Boosting model can correctly identify NSTEMI from UA and non-ACS etiologies using clinical data. The key findings are (1) Our proposed

Table 4
Performance comparison of the boosting ensemble.

Data Processing Levels	Boosting Ensemble Model	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-score (%)	MCC (%)	AUROC (%)
Raw Data	ADB	97.0	41.0	48.0	77.0	43.0	21.0	98.0
	GRB	98.0	50.0	41.0	73.0	44.0	20.0	100.0
	XGB	98.0	52.0	40.0	72.0	43.0	–	100.0
Average		97.7	47.7	43.0	74.0	43.3	13.7	99.3
FS + CSC	ADB	98.0	45.0	45.0	76.0	45.0	24.0	99.0
	GRB	98.0	51.0	41.0	73.0	44.0	20.0	100.0
	XGB	98.0	54.0	39.0	71.0	42.0	–	100.0
Average		98.0	50.0	41.7	73.3	43.7	14.7	99.7
Proposed Framework (FS + CSC+ RS)	ADB	79.0	80.0	80.0	90.0	80.0	71.0	85.0
	GRB	90.0	91.0	91.0	95.0	91.0	86.0	98.0
	XGB	88.0	88.0	88.0	94.0	88.0	83.0	97.0
Average		85.7	86.3	86.3	93.0	86.3	80.0	93.3

framework delivers a good performance level with an AUROC score and F-1 score of 98% and 91%, respectively, and (2) identification of clinical features using BorutaShap. Using BorutaShap, we identified eleven significant predictors: BNP, CHD, CAD, creatinine, glucose, heart attack, heart rate, nephrotic syndrome, RCDW, SBP, and troponin were independently associated with NSTEMI. However, the accuracy, sensitivity, specificity, MCC, and F1-score of the Gradient Boosting model were 90%, 91%, 95%, 86%, and 91%, respectively.

4.2. Clinical implications

In current clinical settings, ACS diagnosis in patients with non-ST elevations is difficult [34]. Incorrect diagnosis of a patient with ACS is associated with significant morbidity and mortality. Consequently, the overdiagnosis of these patients leads to unnecessary testing, eventually leading to a rise in medical care costs. Due to its high sensitivity and precision, our proposed framework can potentially improve the existing standard of care. Current diagnostic tools may fail to meet expectations in diagnosing NSTEMI patients as their diagnostics depend primarily on serum troponin levels and patient histories. We reason that the proposed framework can allow clinicians to improve patient outcomes through more accurate and timely diagnoses. It can also assist physicians with decisions regarding additional treatments and longer lengths of stay. Our proposed framework can offer a chance for financially savvy diagnostics as physicians can rely on a broader range of symptoms to accurately identify patients with NSTEMI, UA, and non-ACS etiologies.

4.3. Interpretation of predictors

In this study, the feature selection process merged data-driven systems and domain knowledge, ensuring a comprehensive selection of clinical variables. From the literature, several risk factors, such as elevated troponin, history of myocardial infarction (i.e., heart attack), and coronary artery disease, are exceptionally associated with NSTEMI [35,36], thus supporting the performance of the proposed framework. This study also showed that troponin strongly indicates NSTEMI cases [37–39]. Additionally, chronic heart disease, elevated creatinine, coronary artery disease, history of myocardial infarction (i.e., heart attack), and high systolic blood pressure were considered solid indicators of NSTEMI case prediction [40–42]. Furthermore, nephrotic syndrome and elevated glucose, BNP, and RCDW are distinguished indicators in our study. Then, a hybrid combination of resampling techniques and cost-sensitive classification addresses the data imbalance. Finally, the ensemble model showed high accuracy, sensitivity, and precision. The models also exhibited extraordinary ingenuity in classifying NSTEMI from UA and non-ACS etiologies.

4.4. Strengths

Our findings show that ML can effectively diagnose and distinguish between NSTEMI, UA, and noncardiac chest pain given in the clinical data. The proposed framework can accurately classify these diseases and identify the top features that significantly contribute to the classification process. As per a review of the literature, this is the first study to develop a support framework to diagnose NSTEMI, UA, and non-ACS etiologies based on clinical data alone. Though our proposed framework shows high accuracy, precision, sensitivity, and AUROC scores, incorporating clinical notes (e.g., treating physician's comments) may further improve the performance of our proposed framework.

4.5. Limitation

In addition, the proposed framework has not been implemented in actual clinical practice yet. The potential variable will likely upgrade clinical dynamics and improve targeted patient wellbeing.

5. Conclusion

Our findings support that our proposed model may enable us to identify significant variables and diagnose NSTEMI, UA, and non-ACS with high accuracy and sensitivity. These outcomes will likely change the way ACS patients are currently diagnosed and will aid in minimizing the risk attributed to the development of life-threatening STEMI. There is an extremely high risk of a major adverse cardiovascular event in wrongly diagnosed stable NSTEMI patients. Early diagnosis of stable NSTEMI and UA has a massive clinical incentive for the choices involved in managing these diseases (such as an early invasive strategy like angiography and cardiac revascularization with percutaneous coronary intervention). Our study uses single-center data to develop a diagnostic decision support tool. Nonetheless, this proposed tool would help drive toward customized medicine and better risk management for NSTEMI, UA, and non-ACS etiologies patients with clinical concerns for ACS.

For our future research, we plan on spatiotemporally validating this on external data to address the generalizability of our proposed framework. Also, we would like to see what effect clinical narratives from physicians' comments have on our classification model. Furthermore, other research areas may include better application of image data obtained during laboratory visits to improve decision support for these groups of ACS patients.

Declaration of Competing Interest

The authors have no conflict of interest in submitting this paper.

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